

NIRAV SURABHI

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Purdue mechanical engineering student and first-author IEEE researcher who ships production software: two native iOS apps on the App Store, real-time voice-AI backends, and instrumented hardware with measured results.

EDUCATION

Purdue University — B.S. Mechanical Engineering (First-Year Engineering), West Lafayette, IN Expected May 2030
Coursework: ENGR 133, MA 265 Linear Algebra, PHYS 172 Modern Mechanics · **Activities:** Purdue Formula SAE
Honors: Presidential Volunteer Service Award, Silver · Mathematics Scholar, Central Virginia Governor's School · Captain, Science Olympiad · 1st place, Aguado Guitar Competition

EXPERIENCE

SpareMe — Founder & Lead Engineer · getspareme.com Jul 2026 – Present

- Shipped an iPhone app to the App Store that delegates phone calls to an AI agent: it dials U.S. businesses, works phone trees, waits on hold, and returns a written outcome; users watch a live transcript and can type instructions, listen in, or take over mid-call.
- Built the SwiftUI client and Node/TypeScript backend on Twilio ConversationRelay (real-time speech) and OpenAI (reasoning), with verified caller ID, encrypted PostgreSQL case state, APNs, StoreKit-funded per-minute billing with signed transaction verification, and fail-closed kill switches and rate limits.

Khali — Co-Founder & Lead Engineer · khali.info 2025 – Present

- Shipped a native SwiftUI iOS app and an Expo / React Native app to the App Store for a short-form news feed that pairs opposing sources one claim at a time; applied to **Y Combinator** Spring 2026.
- Built the Node + Python pipeline that ingests sources, clusters events into stories, generates clips and narration, and ranks the feed, plus an OpenAI assistant that surfaces the evidence behind each claim.

I3 Simulations — CAD Intern Jun 2025 – May 2026

- Produced 3D models and haptic-feedback prototypes for medical-training simulators.

Ember Learning — AI Engineer & Software Intern (formerly DeAP Learning Labs) Feb 2021 – 2022

- Wrote the LLM prompts behind AP-tutor chatbots, built automated grading of AP DBQ/FRQ essays against the official rubrics, and indexed lecture transcripts in a vector database for timestamp search; the team secured a **Microsoft and OpenAI grant** and reached a \$1M+ valuation.

RESEARCH & PUBLICATIONS

Passive Vortex Propulsion in Air — First Author, published on IEEE Xplore · doc. 11567580 2024 – 2025

- Built a wind tunnel and test rig with a soldered, calibrated Arduino load-cell system; measured **3.1 N mean net thrust over 20 trials ($p < 0.001$)**, the first experimental evidence of passive vortex propulsion in air.
- Compared flat and NACA 0024/0012/0009 fins with ANOVA and Dunn's test in Python plus CFD in Fusion 360; won the **AIAA Aerospace Award** at the Regional Science & Engineering Fair (invitation to NASA Goddard) and presented at EngiTek 2025.

PROJECTS

FTC Robotics, Absolute Zero #12096 — Captain & Lead Programmer 2022 – 2026

- Led the team for **four seasons**; designed and built every mechanism in Fusion 360 and wrote the autonomous and tele-op code in Java (PID control, sensor fusion, odometry); advanced to the Chesapeake regional championship.

Oh — Creator, open source · github.com/21J3phy/Oh 2026

- Built shared memory for AI coding agents: hook-driven capture of Claude Code and Codex sessions, local secret scrubbing, pgvector retrieval, and an MCP tool that answers why code is the way it is without interrupting its author.

Motor-Based Resistance Trainer — Battery-powered, no-weights strength trainer 2025

- Built three prototypes (eddy-current, chain-driven, geared) of a portable strength trainer; a 188:1 geared motor under Arduino PWM control produced a linear, repeatable force curve ($R^2 = 0.993$) at a \$215 build cost.

SKILLS

Languages: Python, Java, C++, Swift, TypeScript, SQL

Software: SwiftUI, React Native, Next.js, Node.js, FastAPI, PostgreSQL, Supabase, WebRTC, Twilio, StoreKit, Git, CI/CD

AI / ML: PyTorch, computer vision, LLM agents and tool use, vector search (pgvector), prompt design, evaluation

Hardware: Fusion 360 (CAD, FEA, CFD), 3D printing, Arduino, load cells, soldering, PID / PWM control, sensor fusion